

A REVIEW OF USER ROLES IN COLLABORATIVE VR MUSIC-MAKING ENVIRONMENTS: TOWARDS ROLE DESIGN GUIDELINES WITH WAM JAM PARTY AS A CASE STUDY

Anonymous Author(s)
Anonymous Institution
anonymous@domain.tld

ABSTRACT

Collaborative music making (CMM) in virtual reality (VR) enables embodied, spatial, and socially situated forms of musical interaction. However, when multiple users manipulate shared musical artifacts (e.g., patches, timelines, or parameter sets), concurrent actions can lead to interruptions and conflicting edits that disrupt creative flow.

This paper reviews how user roles are addressed in collaborative VR music-making systems. We examine roles both as musical or social participation (e.g., performer, tutor, audience) and as authority over shared artifacts, determining who can modify what and under which conditions. Across the reviewed works, we identify recurring coordination mechanisms such as personal territories, awareness cues, and rule-based interaction constraints, as well as design tensions including symmetry vs. expertise and freedom vs. protection.

These observations lead to a set of design guidelines that we examine through the case study of *WAM Jam Party*, a node-based collaborative VR patching environment based on Web Audio Modules (WAMs). We discuss how configurable roles and permission schemes could support collaborative composition while preserving creative flow in immersive music-making environments.

Keywords: collaborative music making, virtual reality, user roles, permissions, conflicts, shared virtual environments, computer supported collaborative work, web audio modules

1. INTRODUCTION

Context and motivation: overlapping modifications made by different users are a well-known challenge in computer-supported collaborative work (CSCW), particularly in shared editing systems where concurrent actions may lead to conflicts or inconsistent states [16, 20]. Similar issues appear in Collaborative Music Making (CMM) systems, where multiple participants manipulate shared musical structures or parameters in real time. In such contexts, conflicting actions can create friction in the creative process or produce unintended sonic results that disrupt group flow. The *reacTable*, for instance, illustrates how simultaneous manipulation of a tangible shared interface can lead to conflicting user intentions during performance [21].

In virtual environments, these issues take the form of concurrent edits that must be either managed at the system level or prevented through interaction design. Specifically, Virtual Reality (VR) introduces Shared Virtual Environments (SVEs) where users can meet up and interact through embodied and continuous

interactions as opposed to traditional media which rely on discrete event-based input (e.g. mouse clicks). While these enable strong social presence and shared performance, this real-time gestural interaction allows multiple users to act on the same object at the same time and create friction, making temporal and spatial overlaps more likely and increasing the risk of interference or conflicting modifications between users.

Furthermore, reflections around new audience experiences in VR and the musical metaverse (the collection of music-related SVEs in VR) seem to go in the direction of giving more agency to the spectators and to include them in the experience not only as a passive audience but also as an active part of the experience. Boem et al. frames such VR music experiences as *"a new form of making art, which subvert the conventional roles of the artist fully in charge of the creative process and of a passive audience"* [4] while Bradley still points out that considering the number of HMD owners, some if not most of *"users are likely to attend "at-keyboard" - as viewers, not participants -, watching others experience virtual reality rather than entering it themselves."* [5].

In this context, user's roles and permissions become primary design levers to reduce friction while preserving experimentation, play, and inclusive participation for different usage contexts.

Objectives: this study aims to (i) identify user roles in collaborative musical VR experiences, (ii) analyze roles, permissions, and potential conflict situations requiring explicit design choices, and (iii) derive design implications for structuring role distributions in collaborative VR music systems. More broadly, the goal is to clarify how roles and permission structures can be designed in collaborative VR music systems, with *WAM Jam Party* serving as a concrete application context.

Scope and research questions: to address these objectives, we review prior work across two complementary domains: (a) collaborative VR music instruments and environments, and (b) related collaborative music and HCI systems that address role management through mechanisms such as privacy and awareness, territoriality, asymmetric permissions, and coordination under latency. Based on this review, the study addresses the following research questions:

RQ1. What role concepts recur in collaborative VR music systems?

RQ2. Which coordination and conflict resolution mechanisms (e.g., personal space, locking, turn-taking, moderation) are reported, and how do they support creative flow while preserving collaboration?

RQ3. How can these lessons be translated into configurable role distributions and design guidelines across different CMM contexts (free jam, lesson, performance, game)?

Here, *free jam* denotes unstructured co-creation, *lesson* denotes teacher-led practice with explicit authority, *performance* denotes audience-facing sessions with show control, and *game* denotes rule-driven collaboration or competition.

We aim to account for different types of CMM environments and derive guidelines applicable for a broad range of systems, while focusing our analysis primarily on applications similar to WAM Jam Party.

Target system: WAM Jam Party [11] is a collaborative VR environment in which multiple users assemble and manipulate modular audio systems in real time. Participants create *music installations* by instantiating Web Audio Modules (WAM) plugins [10] from a *virtual shop* and connecting them into audio graphs composed of note generators, instruments, audio effects, and spatialized outputs. Each plugin appears as an interactive 3D object that can be positioned, connected, and edited in space. A typical installation consists of note generators driving instruments, routed through effects and finally connected to spatialized audio outputs placed in the virtual environment. By arranging these components in 3D, participants collaboratively shape both the musical structure and its spatial rendering. Users interact through head-mounted displays (HMDs) and handheld controllers. Nodes can be grabbed, moved, connected, and edited in 3D, while parameters are adjusted through embodied gestures or dedicated interfaces. Some modules expose specialized interfaces such as a piano-roll editor for note sequencing or embodied VR musical instruments (VRMIs) such as a virtual drum kit played with drumsticks. These interaction models build on recent work on immersive WAM-based environments and interaction affordances [12, 11, 25].

All participants share the same musical environment: plugin states, parameter values, musical patterns, tempo, and spatial configurations are synchronized several times per second. However, the current prototype does not implement explicit roles, locking, or ownership mechanisms. As a result, several users may simultaneously manipulate the same musical artifact. While this openness encourages experimentation and improvisation, it can also disrupt collaborative flow, for example when the piano-roll editor records drum patterns played by one user while another participant edits the same sequence.

An open-source WebXR prototype is publicly available¹ and actively maintained², providing a concrete testbed for experimenting with role distributions and permission mechanisms in collaborative VR music systems.

Contributions: the proposed study presents a structured review of user roles in collaborative VR music-making environments and derives design guidelines for CMM SVEs. It further explores these insights through a case study of WAM Jam Party and proposes additional role-related mechanisms to leverage as well as a configurable role-based permission model supported by an open-source WebXR prototype.

This paper is structured as follows. Section 2 describes the materials and methods used to construct and analyze the literature corpus. Section 3 presents the results of the state-of-the-art review and exposes guidelines derived from its synthesis. Section 4 investigates these guidelines implications for WAM Jam Party as a case study. Section 5 discusses various aspects of this work. Finally, Section 6 concludes and outlines directions for future work.

¹ <https://wamjamparty.i3s.univ-cotedazur.fr/>

² <https://github.com/doriangirard9/musical-multiverse-vr>

2. CORPUS CONSTRUCTION, DATA EXTRACTION AND SELECTION FLOW

The corpus was constructed through a **quasi-systematic review**, combining structured search and screening steps, without following the full protocol of a formal systematic review. We started from **3 seed articles** (see the first 3 articles in Table 1), then expanded the corpus using three complementary strategies:

Keyword search: on Google Scholar combining role/permission terms with music/sound-experience terms and VR/virtual-reality terms, using three progressively broader Boolean queries. The final one resulting from the three iterations was the following:

```
("multi-user" OR "multiplayer" OR
"multiple users" OR "collaboration") AND
("role" OR "can modify" OR "can alter"
OR "authorizations" OR "privileges") AND
("music" OR "musical" OR ("interactive"
AND "sound" AND "experience")) AND
("VR" OR "virtual reality" OR "virtual
installation")
```

Backward snowballing: by consulting references cited in the most relevant papers.

Forward snowballing: by tracking papers citing the already-listed papers.

We included papers that explicitly discuss:

- collaborative VR music-making (shared virtual musical instruments, shared VR music spaces),
- role asymmetries or permission structures,
- privacy/awareness or territoriality mechanisms,
- VR-specific constraints (embodied interaction, spatial audio, social presence, spectatorship).

Additionally, some references have been added to the corpus during the reviewing process to get a more exhaustive overview by including relevant industry products [26, 27]. We excluded papers that are non-collaborative, purely technical without interaction design implications, or do not provide interpretable evidence about roles or coordination. For each paper we extracted: system type, interaction modality, explicit and implicit role definitions, conflict/coordination mechanisms, and reported benefits or limitations relevant to creative flow. The overall review process is summarized in Figure 1.

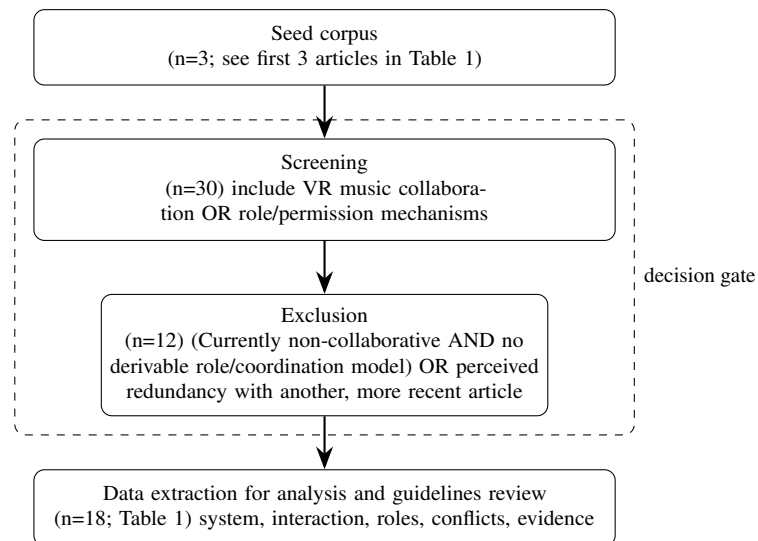


Figure 1. Review pipeline used for this paper selection.

3. STATE OF THE ART REVIEW: RESULTS AND SYNTHESIS

3.1. Corpus characterization

Table 1 summarizes the core and extended works analyzed in order to conduct this review. The final synthesis corpus comprises 16 papers spanning the period 2002–2025 and 2 industrial solutions from 2022 and maintained to this day, reflecting both early collaborative music interfaces and recent immersive VR environments. The corpus includes a mix of VR-native CMM systems and adjacent work on CSCW, mutual engagement and non-VR music applications when they provided transferable role design or coordination mechanisms. Approximately half of the papers focus explicitly on immersive virtual reality environments, shared virtual instruments, or metaverse-oriented music systems (e.g., [3, 28, 1]), while the remainder provide transferable concepts from non-VR collaborative music systems, CSCW research, or web-based collaborative audio platforms (e.g., [20, 7, 2]). These non-VR systems remain relevant because they address fundamental coordination challenges that persist in VR contexts such as awareness, ownership, and concurrent editing. The reviewed systems cover a range of collaborative configurations, including shared virtual musical instruments, distributed networked music environments, browser-based collaborative patching systems, immersive performance exhibits, and educational applications. Several papers report user studies (e.g., [3, 24]), while others present specific SVEs and toolkits (e.g., [14, 15]) or constitute literature reviews around adjacent topics [1, 2]. Together, these works reveal recurring patterns in role distribution and coordination mechanisms that are relevant for the design implications discussed later in the paper.

The following sections analyze these patterns in greater detail, focusing first on how roles are conceptualized in collaborative VR music systems, before examining the coordination mechanisms that regulate interaction between participants.

3.2. Role characterization in collaborative VR music systems

Across the reviewed corpus, two complementary perspectives emerge for describing user roles in collaborative VR music systems. 1) A user role can be understood as a **musical/social role** (what the person is *there to do*). Examples include performer, conductor or bandleader, tutor, learner, active or passive spectator, or organizer. 2) It can also be understood as an **authority/interaction role** (what the person is *allowed to change*). Examples include owner (full control), editor (can modify shared artifacts), contributor (can add but not overwrite), observer (read-only), or moderator (can modify other users' access rights).

These roles are closely linked to permission structures (e.g., adding, deleting, or rewiring nodes, managing access), their scope (own, shared, or global), and to the coordination mechanisms users mobilize to synchronize or communicate with each other (e.g., cues, feedback). This distinction helps avoid a common pitfall: a "performer" or "conductor" may still require restricted authority, while a "learner" may need elevated permissions depending on the context. In many cases, identifying these aspects in prior work requires examining experimental setups and the types of control granted to users, particularly when roles and permissions are not explicitly described.

The following sections examine how these role configurations appear across different collaborative contexts and how they relate to group flow and mutual engagement.

3.3. Multi-role perspectives in different collaboration contexts

Several use cases appear across the reviewed works. While some contexts may overlap depending on the situation (e.g., a user may learn within a gamified environment), we identified context-specific implications that should be considered when approaching multi-role design in CMM SVEs. The following subsections examine these contexts in more detail and present the role and permission mechanisms reported in the referenced articles.

3.3.1. Learning, lessons and accessibility

The benefits of collaborative music-making in virtual environments for supporting music learning (both theory and practice) and social interaction have been widely studied, including in industry platforms such as Patchworld or Fortnite [6]. Fanani et al.'s work on a rule-based learning application showed such benefits by reporting an evaluation made with two experienced musicians recruited to train two novice groups. The first group of users played inside a virtual instructor-led music game between training sessions and reached an *"excellent skill scores"* faster than the second control group who didn't [17]. Collaborative aspects in CMM applications can set the ground for new kinds of lessons inside virtual environments where the role of the teacher impersonated by a user could emerge.

Early work on collaborative musical interfaces emphasizes that technology should primarily support social interaction rather than dominate the experience [2]. Many systems therefore simplify musical control so that novice users can participate without needing to master musical structures such as scales or harmony. This design strategy is particularly relevant for public installations and walk-up-and-play scenarios, where interaction must remain immediately accessible.

Sound Blocks VR similarly foregrounds accessibility considering motor impairments in VR instrument design, presenting an Accessible Digital Musical Instrument (ADMI) that is *"also suitable for many other applications, including music education in young learners and new forms of expressivity"* [19]. This collaborative instrument is fully compatible with controllers as well as hand-tracking (preferred by motor-impaired users) and allows for the use of *"presets for musical scales"* as well as advanced parameter under a button menu. Thanks to this instrument design, an *"impaired musician can participate and even become the soloist at the center of the performance"*, revealing that some musical **entire roles** aren't accessible to all users depending on the related instruments designs. Identifying accessibility barriers in role and interaction designs should remain a top priority to include the broadest audience possible while enabling collaboration and expressivity.

The takeaway from this first use context of CMMs is that a primary aspect in role design should be to ensure accessibility and accommodate for novice or expert profiles of players, as well as considering emerging roles such as a Teacher for lessons-oriented applications. Several role-related game mechanisms can also improve the learning context and are addressed in section 3.3.3.

3.3.2. Scenic and organizational roles in performances and exhibits contexts

Azevedo et al. reviewed 45 works on immersive public installations and conducted a workshop with designers and event organizers [1]. They identified several recurring roles in specific frames inside and around the installations in their synthesis. Their

Paper	Year	Venue / Editor	Type	VR?	Interaction / context	Role focus	Relevance to the topic / WAM Jam Party
[1]	2025	ACM IMX	Conf. paper	Yes	Immersive public installations	Multi-role UX (participants, staff, organizers)	Review on multiple public VR installation, role taxonomy accounting for organizational roles
[3]	2024	NIME	Conf. paper	Yes	Shared VRMIs ; paired user study	Symmetric vs. asymmetric roles	Role confusion when tasks are decoupled; parameter-split control
[28]	2024	NIME	Conf. paper	Yes	Multiplayer VR game; 14 users (Gamelan Land)	Real-world role hierarchy; cultural preservation	Role structure from traditional music communities, management roles
[24]	2021	Audio Mostly	Conf. paper	Yes	SVE music creation; 26 user pairs, 4 configurations (LeMo)	Personal space design; acoustic attenuation	Personal and public territories: Progressive boundaries through augmented attenuation preferred to binary isolation (isolated / not isolated)
[9]	2025	Web Audio Conf.	Conf. paper	No	WAM web-based CMM (Sequencer Party)	Collaboration primitives	WAM-oriented workflow (basis upon which WAM Jam was designed); real-time sync constraints
[18]	2010	NIME	Article	No	Co-located CMM; public + personal spaces	Privacy/awareness balance	Empirical support for private workspaces before publishing
[19]	2025	Springer	Book chapter	Yes	Accessible hand-tracked VRMI (Sound Blocks VR)	Accessibility for novice musicians	Role simplification for non-expert users
[7]	2012	Behav. & IT	Journal	No	Collaborative creativity	Mutual engagement; group flow	Defines engagement signals useful for role/awareness design
[20]	2002	CSCW	Journal	No	Real-time groupware	Workspace awareness framework	Provides vocabulary for awareness cues in shared editing
[8]	2004	NIME	Article	No	Remote music collaboration (Daisyphone)	Shared control and coordination	Early evidence for remote coordination challenges
[13]	2016	NIME	Article	No	Networked virtual environment (Monad)	Game mechanics	Role structuring via constraints, scores
[14]	2020	NIME	Article	Yes	Cross-platform/cross-reality sonic design tool (INVISO)	Novice/expert workflows	Supports role differentiation across devices
[4]	2023	IS2	Conf. paper	Yes	Web-based CMM SVE (Musical Metaverse Playgrounds)	Shared creation contexts	Highlights browser-based shared creation spaces
[15]	2023	JAES	Journal	Yes	Web-based metaverse performance toolbox (Orchestra)	Performance roles	Operational separation in live music contexts
[2]	2003	NIME	Article	No	Contexts of CMM	Restricted control for accessibility	Motivates simplified roles for non-musicians
[17]	2020	iJET	Article	No	Music learning application	Guided learning for novices	Virtual instructor system role
[26]	2022	PatchXR AG	Software	Yes	Industry VR CMM SVE (Patchworld)	Role-based Access Control	Industry VR music experience with world ownership and permissions
[27]	2022	Fast Travel Games	Software	Yes	VR instrument performance environment (Virtuoso)	Contributor-based collaboration (asynchronous)	Industry VR music-making environment; supports community contributions to songs

Table 1. Core and extended corpus for role/permission design in collaborative VR music systems

analysis found several layers of participation surrounding immersive installations, ranging from peripheral spectators and queued participants to operational staff (operators, technicians) and fully immersed participants or performers. This shows that role design must extend beyond end-users to include facilitators and operational staff when there might be one. Additionally, for exhibits happening fully inside VR experiences (imagining several stages inside a shared space), many roles described here in the real world could be ported to the virtual space inside of HMDs (e.g. with passerbys walking through the different installations), making this taxonomy a precious resource to identify emerging roles in SVEs.

In parallel, *Orchestra* [15] provides a toolbox for live audio-visual performances in web-based metaverse contexts based on AFrame/Networked-AFrame³ (NAF) that might be approached as conventional scenic experiences (i.e. with a stage or several installations, lighting, effects...), furthermore setting the ground for operational roles such as scene management and show control.

³ a web multi-user VR framework

The toolbox integrates several components

- A component allowing for speech or acoustic interactions (such as applause) broadcast to other users.
- Components integrating Puredata extended reality (PdXR) patches fully compatible with collaboration, making it "*possible to perform together on virtual instruments or to create interactive virtual installations*".
- Live Coding Components integrating three live coding languages falling into the category of Esolangs (languages with a reduced set of instructions for "*conceptual, creative, and sometimes artistic [...] designs*" rather than general-purpose use) enabling live-coding in VR through virtual keyboards thanks to their simplified design: Bytebeat, StackBeat, and IBNIZ.

Collaboration examples in the article only show two users performing on live coding terminals but they don't appear to be modifying the same source code in the figure shown (the article mentions the source code of components is shared amongst all users but no details are provided about concurrent code modifications). Meanwhile user roles aren't explicitly discussed in the

article, this work underlines a diversity of use cases that might require specific responsibilities to fit in a performance, giving further relevance to a role distribution for performance-oriented CMMs following a detailed structure as the one from Azevedo et al.'s study.

Including the audience inside performances can also be achieved with new interactions designed for audience-related roles. Blaine and Fels highlighted several instruments as soon as 2003 relied on design strategies to foster a sense of community by encouraging movements such as dancing with strangers [2]. SVEs offer new meeting spaces for fully dematerialized performances which could include such interactions to act upon the music produced by performers or add some interactive aspects to usually passive audience roles.

These observations highlight that performance-oriented CMM systems can involve an equal if not broader range of roles than traditional musical ensembles. In addition to performers, roles such as operators, technicians, or facilitators may contribute to managing the experience and assisting participants. Role design therefore extends beyond musical interaction to include organizational and operational responsibilities.

3.3.3. Free play / Game mechanics: reward system, roleplay & virtual roles

Cakmak et al. introduce *Monad*, a collaborative music system that integrates game mechanics through an internal point economy [13]. Players must spend points to perform actions, while contributions can be rewarded either by other players or automatically by the server - which users preferred as it didn't require them to interrupt their work to reward other players. Their evaluation shows that structured constraints can stabilize collaboration but may also influence the musical outcome and interaction dynamics.

Syukur et al. document the role hierarchy of Gamelan (Indonesian music ensembles) studios - owners, managers, instructors, musicians, singers, visitors and clients - in which the total number of instrument ranges from 30-60 and above. This hierarchy was translated into a multiplayer VR game designed for users to play on a long-term basis called *Gamelan Land* [28]. In the experiment, active participants were limited to the instructor, musician, singer and audience while the owner and manager's roles are simplified to scheduling practice and paying studio members and performed by the system like in *Monad*'s second setting. Similarly, the story's narration revolves around the activity of collecting points. Although the game avoids conflicts by assigning one instrument and role per user, the real-world hierarchy shows that traditional ensembles already rely on structured role distributions, making role design a part of cultural preservation as well as a lever to reduce friction.

In addition to the actual user roles, non-playable characters (NPC) virtual roles can also enhance the user experience, for instance by accelerating novice players onboarding through tutorials, whose importance has been pointed out by several studies reported in Azevedo's et al work [1]. This insight is highlighted in Fanani's et al. work where a virtual instructor is implemented inside a Gamelan learning game to teach players how to play Gamelan instruments [17]. This role is fulfilled by the system which runs a model adapting to the player's level to give him or her feedback depending on the system's expectations. Considering Nintendo *Wii Fit* virtual instructor's popularity as of today, virtual roles should be considered carefully to allow for user's immersion and to fulfill specific goals as they can provide them with the drive needed to partake in the experience, or constitute

an obstacle if they are not deemed convincing enough.

While these examples highlight how roles emerge from different collaboration contexts, they do not fully explain how access to shared musical artifacts is technically regulated. The following section therefore examines mechanisms used to structure authority and interaction between participants.

3.4. Role-based access control (RBAC) as a conflict prevention and experience design lever

While the previous section examined role distributions across contexts, the following subsections analyze mechanisms used to regulate access and coordination between roles.

3.4.1. Symmetric vs asymmetric roles in shared VMIs

Boem and Turchet introduce the *Musical Metaverse Playgrounds*, a web-based VR application hosting shared virtual sonic experiences [4]. Two "playgrounds" are presented. In both of them users can move through the environment, communicate through voice speech and interact with 3D objects in the scene through gestures. In the first playground, users can generate three dimensional synthesizers and play them simultaneously, while the latter focuses on multimodal interactions by enabling sound-generation accounting for the users microphone inputs visually represented through a series of animated vertical bars. Additional 3D visualizations which can only be controlled by their creators "to avoid confusion" can be created by the users. 8 users were paired with an examiner to conduct experiments in both playgrounds through a thinking-aloud protocol that was followed by an unstructured interview. These experiments give interesting insights through user's feedback considering roles and permissions as perceived by the users. Notably, in the first playground, three participants "found confusing that any connected user could modify the sound generators, even if they weren't the creators", pointing out that "such type of collaborative activity (i.e., changing together the parameters of a sound generator) is dependent on the type of community involved and on how much a person will trust the other users". Consequently, "They highlighted that it should be important to set some boundaries" and suggested the implementation of a mechanism to "share the ownership of the generators they have created".

Shared VMIs also raise a fundamental shared-control problem mentioned in this paper's introduction: performers can simultaneously modify parameters of a single interface [3]. Boem et al. compare three VRMIs with distinct role configurations - strongly asymmetric (a player moves the sound source while the other changes the sound it produces), partly asymmetric (identical controls but managing different synthesis parameters of the instrument), and fully symmetric (both users can do the same things) - in a user study with 12 participants. The strongly asymmetric instrument was perceived as confusing ("I do not quite understand what my role is"), partly because one player was in charge of moving the sound source, making the other player unable to hear the spatialized feedback of what he was doing. By contrast, the partly asymmetric design achieved the best usability (SUS) and creativity-support (CSI) scores, and also fostered stronger social presence, suggesting a link between role clarity and collaboration quality. It is also notable that the fully symmetric instrument incited players to coordinate between themselves ("You do the kick drum, I do the snare"), preventing conflicting intentions.

As seen in learning contexts, educational VR music systems might also require asymmetric control to support instruction and assessment or to allow novice and expert players to play together without creating chaos. Boem et al.'s conclusion over the three

types of instruments introduced also go this way, indicating the strongly asymmetric design *"may be better suited for more experienced players. Conversely, systems like those of [the 2 other designs] appear to be more ideal for novices"*. Rule-based learning applications show how structured feedback and constrained interaction can guide novice learning, which translates into teacher-led authority models in VR lessons [17]. A user impersonating the teacher could need to be allowed greater control over the session and manage access for a single or several students to specific sets of nodes or parameters depending on the size of the "class".

In gaming contexts, asymmetry should not introduce imbalance amongst the players. *Monad* players participated in an evaluation combining local and remote players. When discussing the possibility of adding embodied interactions to local players in the physical space available, said local players *"expressed that physical gestures would go against the democratic nature of the performance by way of overpowering the role of the remote players in the audiovisual output"* [13].

3.4.2. Cross-platform role design and parameter-split

INVISIO [14] introduces a cross-platform interface supporting VR and AR for creating virtual sonic environments on desktop or inside of HMDs. While collaboration is only mentioned as a future perspective of the work presented, the cross-platform aspect is interesting to account for such an evolution, as users might access the SVE from different devices depending on their social role and might simultaneously control different parameters of shared audio sources in the scene.

Boem et al.'s study [3] tested such a parameter-split instrument, where one user could move the sound location while the other user changed the sound it emitted. Their study tends to disavow the viability of this solution as one of the user was not able to hear what he was doing due to the other player's manipulations of the localization. We might still wonder if a different design would have made this solution viable (by allowing the user changing the sound to hear his work in a private space for instance). At the same time, they report splitting control over different sound synthesis parameters of the instrument is well perceived by users, emphasizing the potential of sharing controls if done thoughtfully.

The cross-platform aspects also resonates with Azevedo et al.'s categorization of the Technician role [1], who monitors the VR experience from a desktop environment. As part of the scenic experience, a captation of performances could be done by technicians on desktop in real-time through a set of cameras placed in advance in the scene or controlled by spectators. The output could be displayed on a flat-surface screen to enable a broader audience outside of HMDs to attend the experience, as well as opening the possibility for some users to contribute from an actual keyboard, enabling the use of traditional live coding languages in addition to Esolangs for toolbox's such as Orchestra [15], or the appearance of additional roles compatible from desktop access such as moderators.

3.4.3. Session sharing in current CMM spaces

Patchworld [26] and Virtuoso [27] are the two main industry references for VR CMM SVEs while Sequencer Party [9] is a node-based, web-based CMM environment and the "ancestor" of WAM Jam Party. All three enable collaboration between users and implement a declination of role-based access control. Designed as fully functional solutions operating since several years, insight can be gained by analyzing their permission models.

In PatchWorld, users can create public or private "worlds" that they own⁴. Public worlds can be seen by anyone. Private worlds can be seen if shared with at least the "View" permission level, or if they are published. Both private and public worlds can be shared with 2 additional permission levels: Edit (a designated user can edit) or Open for Edit (anyone can edit). Worlds can also *"be linked together to create interconnected experiences"*. Additionally, during live sessions, the host (creator) can access a Mic/Touch/Edit permission menu listing each users permissions on his wrist⁵, allowing for fine-grained control.

Similarly, in Sequencer Party, *organizers* can create, edit or delete sessions of their own in which they have every right to add, change or remove tracks and assets (nodes). They can invite guests (passive audience) or collaborators to their sessions. Collaborators can be

- **performers**, who *"can manipulate parameters of WAM components but cannot add or delete such components"*.
- or **editors** who have a *"structure control"* and can *"add, delete or rewire component; change plugins and edit generative logic or sequence content"*.

Virtuoso, however, does not currently support real-time collaboration over internet. Users can share their musics in a Community Library, where other users can pick them up and contribute to make "asynchronous" collaborative songs. All players participating are then listed in the song as contributors⁶. Nevertheless, Virtuoso supports local collaboration through Ableton Live, mitigating long-distance delay issues. In such contexts, the first user to connect determines the tempo. Any connected user can modify it, and if several users do it simultaneously, a race occurs and the last one "wins". A user cannot be prevented from modifying the tempo if he wants to⁷. Additionally, extreme tempo changes in opposing direction can make *"the beat timeline diverge - However, Link will always keep the phase of the apps in sync"* and recording a performance can be made by:

- Having each participant record its performance and merge sessions later
- Mixing everyone through a single mixer and record the stereo output
- Alternatively, multi-track record to one of the Live users' computers.

The responsibility of the recording can therefore be split among the different users and require coordination to stitch the pieces together or can be delegated to a single user recording every track.

These systems illustrate how role-based access control can structure collaboration in practice but permissions alone are not sufficient to support smooth collaboration. Participants must also remain aware of each other's actions and coordinate their activities in real time. The next section therefore examines coordination mechanisms supporting mutual engagement and group flow.

3.5. Coordination mechanisms regarding mutual engagement and group flow

3.5.1. Awareness

Permissions alone do not determine the quality of collaboration; it also depends on whether participants experience *mutual en-*

⁴ <https://wiki.patchxr.io/patching/worlds>

⁵ <https://wiki.patchxr.io/general/live-session>

⁶

⁶ <https://virtuoso-vr.com/faq>

⁷ <https://www.ableton.com/fr/live-manual/11/synchronizing-with-link-tempo-follower-and-midi/>

gement, a state of group flow in which users respond to each other's actions and ideas [7].

Prior CSCW works (i.e. by Gutwin and Greenberg) shows that explicit locking mechanisms become less necessary when collaborators are sufficiently aware of each other's actions [20].

Building on this ground, Bryan-Kinns and Hamilton argue that the key features allowing for mutual engagement are the following:

- **Mutual awareness of actions:** Highlighting new contributions, identifying who contributed, where...
- **Shared and consistent representations:** Views on the joint product should be the consistent across users
- **Mutual modifiability:** Being able to modify each other's contributions - Implying an egalitarian approach to role assignment rather than role enforcement.
- **Annotation:** Being able to converse in and around the shared product.

As a result of their study on the *Daisyphone*, a 2004 shared virtual instrument applying these features, they suggest that "*role assignment emerges naturally and does not need to be explicitly built in to the interface*" [8]. Comparing two versions of the daisyphone, the authors report that by simply changing the note persistence to a decaying version, sessions became unstructured and resembled a lot more conventional music making in real life as users had to focus and had a lot less time to communicate, showing that role-design (or its absence) is tightly coupled to the experience goal.

Analyzing their results, they finally argue that mutual modification in CMM contexts is *not as strong an indicator of mutual engagement as coherence and musicality of the final product*. Users who felt the most engaged tended to be the ones with less activity, who modified the contributions of others less and rather focused on their own contributions to the shared piece. This was also facilitated by the presence of annotations and IDs which allowed for less time dedicated to organizing the tasks as users could see what others were contributing and collect information quickly. Authors argue accountability through IDs may also have made the users more cautious of their contributions. Earlier studies on the *Daisyphone* also showed the importance of localization - being able to reference parts of the shared product to discuss them with others or propose revisions - and implemented it through an annotation system using the mouse to draw graphical lines (users can write or circle parts of the composition). This is supplemented by a history system through automatic annotations when notes are set / unset that allows user to see who made which edit and when they made it, providing context without wasting the restricted communication windows.

The importance of communication is pointed out in several other studies with different modalities, from the *Musical Meta-verse Playgrounds* where "*All participants commented positively on the presence of a real-time voice chat system*" to *LeMo* where participants took turns using a virtual pen to write "*I make*" or "*You do*" in the 3D VR space [22] as well as *Monad* which implemented a chat system to compensate for the absence of visual cues band members can usually give each other in real life performances.

Identification of who is contributing has also been a concern for *Monad* with users showing a "*strong preference for a clear representation of each user's presence within the environment*" and "*feeling dissatisfied when their actions were not evident in the audiovisual output*".

3.5.2. Privacy and territories

A key finding is the value of **separating exploration from publication**. Fencott and Bryan-Kinns show that musicians make extensive use of private workspaces when it's available to them [18]. They report users defaulting to a divide-and-conquer approach when it's not available, where each player takes on an informal role (e.g. focusing on bassline or drums). Personal spaces rather let users experiment in private before sharing their work for "*selection, scrutiny and further revision*". In a thorough study, Men et al. test four sound attenuation configurations with 52 users composing music in pairs in *LeMo*, a VR CMM SVE [24]. They revealed adding personal space improves individual creativity, but rigid wall boundaries increase distance between collaborators and reduce group edits. Their key contribution is the concept of **acoustic attenuation as a fluid personal space**: the system gradually attenuates what a user hears from the shared space, enabling continuous adjustment between full isolation and togetherness. They derive five design implications SVEs, with three specifically for CMM SVEs:

- CMM SVEs should provide users with personal spaces
- Augmented acoustic attenuation is an efficient way to do so for audio-related task. Generally rigid walls should be avoided but are still better than no personal space [23] (a rigid space with mobility is better than a fixed one too).
- The level of attenuation defines the level of "personalness", therefore it should be configured accordingly and made configurable for users to adjust.

Awareness and privacy are key factors that will define how users approach the concept of roles. When no formal role distribution is given, users will develop strategies to organize themselves whether it is turn taking or divide-and-conquer. Designing with awareness and privacy principles in mind can reduce the need for a formal role distribution system as users naturally tend to focus on their own part of the work when provided with personal spaces and clear cues as to what others are doing. Collaboration primarily occurs through communication which can preempt most of conflicts (users' actions are less likely to overlap). Implementation of such concepts should be prioritized (knowing it will affect how users approach the experience) in order to identify the remaining conflicts that a formal role system could solve if there is still a need for it.

These findings highlight patterns in how collaborative music systems manage roles, permissions, and coordination. The following section synthesizes these observations and derives a set of design guidelines for collaborative VR music-making environments.

3.6. Cross-paper synthesis: role design guidelines

Across the reviewed works, several recurring patterns appear regarding role distribution, coordination mechanisms, and conflict management. Instead of converging toward a single role model, the corpus points to a set of recurring *design guidelines* for structuring roles and permissions in collaborative CMM SVEs.

G1 – Consider the broader ecosystem of roles: Collaborative VR music systems involve more actors than performers alone. Immersive installations highlight roles such as facilitators, technicians, or operators, as well as multiple forms of audience participation ranging from passive spectators to active participants. Educational systems further introduce teacher–learner dynamics while gaming contexts show virtual roles can complement user roles too. Role design should therefore account for this broader ecosystem surrounding the musical activity.

G2 – Support diverse expertise levels: Many systems aim to accommodate both novice and expert users. Accessible interaction design can support novice participation while preserving expressive potential for experienced musicians. In such contexts, configurable role asymmetry can help structure collaboration and avoid chaos or errors, for instance in learning or performance preparation scenarios.

G3 – Use explicit roles only when necessary: Explicit role assignment is not always required and can alter the user experience. In several systems, users naturally adopt informal coordination strategies such as turn-taking or task division when roles are not predefined [22]. However, explicit roles may still be useful in contexts where authority must be enforced or to support roleplaying [28].

G4 – Prioritize workspace awareness: Workspace awareness is a key mechanism for preventing conflicting actions. Participants benefit from cues indicating *who is interacting with what, where, and when*, supported by communication channels such as speech, chat, or visual annotations [20]. Visibility of contributions is particularly important, as users report frustration when their actions are not clearly identifiable. Roles should also remain understandable to participants to avoid confusion [3].

G5 – Provide personal territories for experimentation: Musicians frequently prefer to experiment before exposing their work to the group. Personal territories allow the separation of exploration and shared production. In VR environments, augmented acoustic attenuation has been shown to provide a flexible mechanism for modulating privacy between shared and isolated listening conditions [24], letting users adjust their level of "personalness".

G6 – Keep ownership and editing policies configurable: Different systems adopt different approaches to artifact ownership. Some rely on fully shared editing coordinated socially, while others introduce explicit ownership or permission structures. User studies indicate that participants sometimes expect control over artifacts they create [4], suggesting that editing and ownership policies should remain configurable to balance creative freedom and protection from destructive edits.

Looking back at these guidelines, several recurring design tensions can appear in collaborative CMM systems, notably *symmetry vs. expertise*, *freedom vs. protection*, and *awareness vs. privacy*. The next section examines how to address these tensions.

4. CASE STUDY: WAM JAM PARTY

This section explores how the guidelines derived from the literature can inform role design in WAM Jam Party. The following subsections discuss potential mechanisms and role configurations inspired by the review.

The need to work on role distribution in WAM Jam Party came from continued development on the app, with next steps requiring careful thinking before starting to implement specific behaviors based on use contexts as well as persistence for multi-sessions management to make the application relevant for users to use in autonomy beyond being a research subject. Informal testing of WAM Jam Party and Sequencer Party revealed several conflict situations caused by unrestricted permissions and limited awareness cues. During a class presentation, for instance, a connected teacher adjusted track volumes and panning to improve his personal listening mix, unaware that these changes were synchronized across the shared session. Students therefore observed unexpected modifications in their patches. In another case, a user recording a drum pattern through the shared piano roll and

was interrupted by another participant simultaneously editing the same interface.

This motivates explicit role mechanisms, but the literature showed it might not be necessary if sufficient awareness cues and communication means are provided depending on the context. For WAM Jam Party, being open-ended towards multiple use contexts, it might be sufficient to rely on conflict avoidance mechanisms alone although explicit role systems may still be required in some contexts. Obviously, the best solution would be to make the settings totally configurable by users to let them customize their experience, but the balance between complexity and customizability has not been clearly identified.

In the following subsections we will detail each derived guideline implications, discuss additional mechanisms that can be leveraged and propose an adjustable permission matrix for retained roles.

4.1. Applying synthesis guidelines to WAM Jam Party

G1 – Consider the broader ecosystem of roles: : role design should account for organisational roles in performance contexts, to dissociate scene setup and performance recordings made by technicians and performer controls. Considering the node-based nature of WAM Jam Party, there is little need for further specification as the type of node used by a performer will determine his temporary musical role if he was in a traditional band (singer, drummer...).

G2 – Support diverse expertise levels: tutorials are important to support onboarding and reduce early interaction errors. Virtual instructors could be envisioned to do so, but basic tutorials can preempt chaotic situations rather than enforcing strict limitations resulting in user frustrations. Interactions should be designed to favor accessibility which will reduce friction for every user profile, and a specific novice profile might be added to allow for users who select it to use pre-built audio chain parts considering the node-based aspect of WAM Jam Party. The addition of such a profile remains non-necessary if such presets are accessible for every users as even experts could decide to use them.

G3 – Use explicit roles only when necessary: public sessions should default to symmetric control, allowing users to self-organize unless explicit roles are required. Users should nevertheless be allowed to save sessions and become owners of their iteration. Sessions could then be shared for iterative improvements through a similar system as that of Patchworld or Sequencer Party.

G4 – Prioritize workspace awareness: Awareness of participants' identity, contributions, and current actions is essential to reduce reliance on complex permission systems. This can be supported through cues such as highlighting new contributions, visually linking users to the nodes they are modifying (e.g., with colored borders), or displaying notifications indicating where sound modifications originate.

Communication mechanisms should also support coordination between users. Proximity voice chat and lightweight annotation tools (e.g., a virtual pen to point at objects or parameters) can facilitate discussion within the shared workspace. Additional cues such as simple avatar gestures (e.g., "help me" or "come here") may further support coordination. Finally, a contribution history should be provided and designed to remain usable within head-mounted displays.

G5 – Provide personal territories for experimentation: WAM Jam Party already implements an augmented sound attenuation system with multiple spatialized outputs. However, the spatial

range of these outputs is currently fixed and should ideally be configurable. Additional listening modes or mechanisms to temporarily group nodes for private monitoring could also be considered. Since outputs are shared, participants may unintentionally interfere with each other's listening setup. Simplifying the creation of private listening contexts would reduce the need to repeatedly rewire nodes to different outputs during iterative work.

Additionally, prototypes for VRMIs such as the virtual drum kit already implement a form of personal territory (gestures remain local until the user records and publishes) [25]. Generalizing this into a *copy/push* model, where any user can audition a personal copy of a subsystem with acoustic privacy before pushing it to the shared session, could significantly reduce accidental disruptions during iterative creation.

G6 – Keep ownership and editing policies configurable: As discussed above, a permission system may be required to address conflict situations that cannot be resolved through awareness or coordination mechanisms alone. Such a system should remain configurable by session owners while avoiding excessive complexity.

Node groups could be associated with simple locking mechanisms when users own the corresponding nodes. Similarly, users performing critical tasks—such as adjusting continuously modifiable parameters or recording patterns with VRMIs—could temporarily lock the relevant parameters or nodes to prevent conflicting edits.

Session sharing mechanisms should also preserve ownership of saved installations, preventing other users from unintentionally altering or destroying existing configurations.

4.2. Additional design mechanisms

Beyond the guidelines discussed above, several complementary mechanisms could further support collaboration in WAM Jam Party and similar CMM SVEs.

Audience participation and lightweight interactions: game-inspired mechanisms may help integrate audiences more actively into collaborative performances. Rather than assigning editing permissions, spectators could interact through lightweight actions that do not interfere with the musical structure (e.g., triggering visual effects, influencing reward systems, or reacting through gestures, competition modes). Such mechanisms could strengthen engagement while preserving the stability of the shared musical environment. Audience-related roles could also benefit from dedicated accessible interactions requiring no permissions to favor inclusion, such as "dancing with others" interactions acting upon the shared musical product.

Cross-platform roles. Because WebXR environments can be accessed from both HMDs and desktop browsers, role-specific interfaces could support different responsibilities. For instance, moderators or technicians might supervise sessions or manage permissions from desktop interfaces, while performers interact within the immersive environment.

Distributed parameter control. Control over musical parameters could also be distributed across participants. Previous studies suggest that such designs can support collaboration when responsibilities remain clear and users can perceive the effects of their actions.

4.3. Proposed role distribution by use context and permission matrix

Table 2 proposes a context-sensitive role–rights matrix for WAM Jam Party, inspired by multi-role UX perspectives [1]. Roles are organized according to typical collaboration contexts (performance, lesson, free jam), and each column indicates whether a

given role can perform a specific action on the shared musical structure. The matrix reflects varying levels of authority, from read-only audience roles to full-control session owners. Intermediate roles such as performer, contributor, or technician represent different degrees of interaction with the shared patch depending on the session context. This matrix is intended as an initial design proposal and will need to be refined through user testing.

Implementation perspectives are inspired by session sharing mechanisms from Patchworld and Sequencer Party. Owners can choose session contexts amongst the one proposed. Sharing for edit will provide users with an intermediate-access role (performer, contributor/player, student) while sharing for viewing will be used to invite audience members. Other roles will be assigned manually by owners and moderators that owner's nominate.

Note that in our design, users who can add nodes implicitly retain the right to edit or delete the nodes they created.

In practice, moderation privileges do not necessarily correspond to a participant's musical role. A moderator may simultaneously act as a performer or contributor during the session. For this reason, we argue that moderation capabilities should be considered as an additional privilege layer rather than a distinct social role. In this model, a participant may combine a base role (e.g., performer, contributor, student) with moderation privileges such as managing access rights or removing disruptive users. This separation avoids unnecessarily promoting moderators to full session owners while still enabling effective session supervision.

We considered further specializing the distribution with sub-roles such as sound engineer (controlling mix, spatialization rather than instruments parameters...) but we preferred to keep the distribution as simple as possible, hence the common owner role for each highest authority level, as teachers or game masters didn't need further specification regarding permissions. Furthermore, the node-based design of WAM Jam Party could have complicated concrete implementations of such a specific subdivision.

Different collaborative contexts motivate different role distributions. Three representative scenarios illustrate how the proposed permission matrix may support typical uses of WAM Jam Party:

Performance preparation: a group of musicians collaboratively builds a musical installation that will later be performed live. During the preparation phase, technicians may add and modify nodes, while a session owner supervises the overall structure. Once the installation stabilizes, performers focus on playing instruments while technicians monitors the scene and can adjust split parameters such as spatialization or mixing parameters.

Music lesson: in an educational context, a teacher guides several students through the construction of a musical patch. Students are allowed to manipulate owned nodes but cannot alter the global structure of the installation. Observers or additional students may join with read-only access to follow the lesson without disrupting the session.

Open collaborative jam: in informal sessions, participants may freely add and modify nodes while self-organizing their musical roles. In this context, default symmetric permissions (represented through the contributor/player role) combined with awareness cues may be sufficient to coordinate interaction without strong authority structures.

5. DISCUSSION

Although this study is based on a limited corpus of VR-first collaborative music systems, it highlights several recurring design patterns regarding roles, territories, and embodied interaction.

Context	Role	Add nodes	Edit params	Rewire/delete	Promote/demote	Record performance	Grant/revoke	Kick/ban
Common	Session owner	■	■	■	■	■	■	■
Common	Moderator	□	□	■	■	□	■	■
Common	Audience	□	□	□	□	□	□	□
Performance	Performer	■	■	□	□	□	□	□
Performance	Technician	■	■	■	□	■	□	□
Game/Free jam	Contributor/Player	■	■	■	□	□	□	□
Lesson	Student	■	□	□	□	□	□	□

Table 2. Adjustable Role-rights matrix for WAM Jam Party session contexts. ■= right granted; □= right not granted.

Rather than proposing a definitive taxonomy, the analysis identifies a set of role-related mechanisms that appear across systems and contexts. These include the emergence of organizational roles around performances, the importance of spatial territories in immersive collaboration, and the role of awareness cues in maintaining group flow.

This territorial dimension is closely related to the awareness and engagement mechanisms described in Section 3.5. Workspace awareness frameworks emphasize that collaboration depends on participants’ ability to understand who is acting on what, and where. In immersive environments, this awareness is mediated through spatial proximity, avatar embodiment, and visible interaction cues. Roles, ownership indicators, and activity feedback therefore contribute not only to conflict prevention but also to the construction of mutual engagement and group flow.

While the guidelines and permission matrix discussed in Sections 3.6 and 4.3 aim to prevent disruptive conflicts, it remains an open question whether all forms of interference should be eliminated. In improvisational contexts, overlapping control and negotiated interaction may sometimes support creative exploration rather than disruption, which highlights a fundamental tension between stability and openness.

More broadly, immersive patch-based systems blur the distinction between instrument and environment. In traditional settings, the instrument is a bounded object with a clear owner. In collaborative VR patching environments, the instrument becomes a shared, evolving structure that multiple users can modify. Roles and permissions therefore operate not only as access control mechanisms but as structuring elements of the instrument itself. From this perspective, designing role systems is inseparable from designing the instrument’s interaction model.

Another aspect that may become relevant as collaborative VR music platforms mature is user authentication and data management. If persistent user profiles and shared sessions are introduced, questions related to identity management, privacy, and data ownership will emerge. Recent work on decoupled VR music systems highlights the importance of separating interaction layers from data storage and identity management to maintain both flexibility and security. Future iterations of WAM Jam Party may therefore need to integrate authentication mechanisms while preserving the openness of collaborative sessions.

Lastly, the availability of an open-source prototype provides an opportunity to investigate these questions empirically. Future work could examine how territorial mechanisms, ownership models, and role configurations affect collaboration dynamics, creative flow, and user experience. In particular, comparing embodied territorial interaction (e.g., instrument-local control such as the drumkit) with explicit permission-based coordination could help clarify how spatial interaction and formal role structures complement or substitute each other.

The design guidelines, additional mechanisms and permission

matrix proposed in Sections 3.6, 4.2 and 4.3 represent an initial operationalization of these principles, but their broader implications for musical interaction remain to be explored empirically.

These observations suggest that roles, territories, and embodied interaction form an interconnected design space. Rather than being treated as purely technical coordination mechanisms, role and permission systems should be understood as central components of interaction design that shape how collaborative musical experience unfolds in immersive environments.

6. CONCLUSION

This review analyzed recurring or derivable role-related assumptions across collaborative VR music systems (**RQ1**) or adjacent CMM systems, and distilled a set of design principles supporting collaboration and creative flow while reducing the need for a formal role distribution system (**RQ2**). Through a case study applied to the open-ended CMM SVE WAM Jam Party (**RQ3**), these principles are applied and complemented by additional mechanisms extrapolated from the review as well as a preliminary permission model proposition adjustable for future improvements.

Beyond its practical implications, this work also highlights the need to further investigate how role and permission models shape musical agency, authorship, and collaborative experience.

The results of this study are currently being implemented in the open-source WebXR prototype of WAM Jam Party. This implementation will serve as a testbed to evaluate the proposed role distributions and permission mechanisms in realistic collaborative music scenarios. One planned scenario involves the preparation of a collective musical performance with approximately ten musician–composer participants. This process will include an initial exploration phase, followed by the collaborative construction of a shared musical setup and subsequent rehearsal sessions of the intended composition. These stages will allow us to observe how role configurations support coordination, creative flow, and conflict management during extended collaborative work. Such experiments will provide empirical feedback to refine the proposed model and assess its applicability to real-world collaborative VR music-making contexts. This scenario is inspired by a prior collaborative experiment conducted with Erasmus Mundus DIGICREA⁸ Master students from University Jean Monnet, France, in December 2025, using the sequencer.party platform, which will serve as a useful reference for analyzing collaborative dynamics in WAM Jam Party.

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⁸ <https://master-digicrea.univ-st-etienne.fr/en/index.html>

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